

Chapter 3 : part 3

① Multivariate Hypergeometric RV

like picking balls from an urn without replacement.

r : different types

n_i : number of i th type selected/picked. present.

x_i : number of i th type picked.

$$P_{X_1, X_2, \dots, X_r} (x_1, \dots, x_r) = \frac{\binom{n_1}{x_1} \binom{n_2}{x_2} \dots \binom{n_r}{x_r}}{\binom{n_1 + n_2 + \dots + n_r}{n}}$$

$\rightarrow n_1 + n_2 + \dots + n_r = n$

② Multinomial Distribution

like picking balls from an urn with replacement

k : no. of diff types

n_i : no. of i th type present

x_i : no. " " " picked.

$$P_{X_1, X_2, \dots, X_k} (x_1, x_2, \dots, x_k) = \frac{n!}{x_1! x_2! \dots x_k!} \times p_1^{x_1} p_2^{x_2} \dots p_k^{x_k}$$

where $p_i = \frac{n_i}{n}$; $x_1 + x_2 + \dots + x_k = n$

Categorical Distribution

Like picking a target ball from an urn.

Bivariate Gaussian (Normal) Distribution

Standard distribution in a bivariate RV.